

STAT/MA 41600

In-Class Problem Set #11: September 18, 2026

Note: Question 1 has a different setup than in Problem Set 10, but Questions 2, 3, 4, 5 are all related to the setup from Problem Set 10.

1. Alice rolls a 6-sided dice (numbered 1 to 6), and Bob rolls a 20-sided dice (numbered 1 to 20).

Let X denote the sum of the two values (that appear on their two dice). What is the expected value of X ? (Hint: You should split X into the sum of two different random variables, one for each of the dice.)

2. Consider a randomly chosen bear family, consisting of 3 bears, namely, a daddy bear, a momma bear, and a baby bear. Each bear has six possible colors, namely, red, orange, yellow, green, blue, purple. If we pick such a bear family randomly, the sample space consists of all ordered 3-tuples whose elements are r, o, y, g, b, p . Assume that all such outcomes are equally likely. Let X denote the number of green bears in a randomly selected bear family, so $0 \leq X \leq 3$.

Let $X_1 = 1$ if the daddy bear is green, and $X_1 = 0$ otherwise.

Let $X_2 = 1$ if the momma bear is green, and $X_2 = 0$ otherwise.

Let $X_3 = 1$ if the baby bear is green, and $X_3 = 0$ otherwise.

Notice that $X = X_1 + X_2 + X_3$. Now find the expected value of X .

3. Suppose that a drawer of t-shirts contains 7 t-shirts from Purdue and 3 t-shirts from IU. Alice randomly selects t-shirts from the drawer, **without replacement**, until the first IU t-shirt is selected. All possible orders of selection are equally likely. Let X denote the number of t-shirts she chooses, until the first IU t-shirt appears (including the draw for the IU t-shirt itself), so $1 \leq X \leq 8$.

Label each of the 7 Purdue t-shirts with the numbers 1 through 7, so that you can keep track of which is which.

Let $X_j = 1$ if the j th labelled Purdue shirt is chosen before the first IU t-shirt (and therefore before all of the IU t-shirts), and $X_j = 0$ otherwise.

Notice that $X = 1 + X_1 + X_2 + \cdots + X_7$, where the “1” is because we always need 1 shirt for the IU t-shirt itself. Now find the expected value of X .

4. During a 5-day week (Monday to Friday), each day is sunny, cloudy, or rainy. If we pick a week randomly, the sample space contains all ordered 5-tuples whose elements are s, c , or r . One outcome in the sample space is (s, s, c, r, s) . Another outcome is (c, r, c, s, s) . Assume that all of the weekly outcomes are equally likely.

Let X denote the number of sunny days during a 5-day weather prediction.

Let $X_j = 1$ if the j th day is predicted to be sunny, and $X_j = 0$ otherwise.

Notice that $X = X_1 + \cdots + X_5$. Now find the expected value of X .

5. Define U, V, W as independent random variables with $P(U = u) = \frac{e^{-1}}{u!}$ for integers $u \geq 0$, $P(V = v) = \frac{e^{-1}}{v!}$ for integers $v \geq 0$, and $P(W = w) = \frac{e^{-1}}{w!}$ for integers $w \geq 0$. Then define X as the sum, i.e., define $X = U + V + W$.

[You do not need this fact to solve this problem, but please take our word for it, that the probability mass function of X is exactly $P(X = x) = \frac{3^x e^{-3}}{x!}$.]

Find the expected value of $X = U + V + W$ by finding $E(U)$ and $E(V)$ and $E(W)$.